LLM 3-oy 11-Dars Report

RLHF usullari qiyosdagilar.

1) Copy of our SFT model (+ training)
2) Our base model (+ training)
3) A bigger open LLM (e.g. Qwen 7B + training)
4) Already-made reward model from huggingface
5) A frontier LLM directly

LMF faqat uning hakami (Reward Model) qanchalik yaxshi bolsa, shuncha yaxshi ishlaydi.

RL da model oziga qaraab yozadi va reward model (hakam) bergan bal/dan organadi.

1-usul (base model) / RLHF vs: reward model = base model (Qwen 0.5 B): NN architecture

Embedding Layer: Kiruvchi belgilarni tokenlarni manbaniki vektorlarga aylantiradi

Backbone (Transformer Qatlamlari): Matnning ma'nosini va kontekstini tushunadigan ortadagi ulkan qatlamlar (Attention, MLP bloklari)

The Head (Modelning Boshi): Eng oxirgi chiqish qatlami. Aynan shu qatlam modelning vazifasini (matn chiqarish yoki ball berish) belgilaydi.

2-usulda: Base model (Qwen 2.5 – 0.5 B)
Reward model + PPO – RLHF v1

1- usul Natijasi
Reward model – Qwen 2.5 – 0.5 B base + trained
RL method – PPO
Training data – 333 pairs
Reward model accuracy – 0.72 (18 real pairs)
Test score – 40/68
Average answer length – 61 characters
Math problem (judge learned)

4-usul: RLHF: SFT model + Skywork (reward model)
Xitoy kompaniyasi ishlab chiqqan
Tayyor o'rgatilgan LLM model faqat
Reward qilishga moslashtirilgan
(Skywork – Reward – Llama – 3.1 – 8B
yoki Skywork – Reward – V2)
API key kerak emas

3-usul: RLHF: SFT model + Frontier
LLM (reward model)
Groq api key
OpenAI key
Gemini key
Claude key

RLHF ni rivojlantirish metodlari.

- More human data
- AI feedback (RLAIF)
- Verifiable rewards (RLVR)
- Bigger, better judge
- Many judges (ensemble)
- Step-by-step reward
- Length control
- KL penalty
- GRPO / RLOO
- Best of N (rejection sampling)
- Iterative RLHF
- DPO instead of PPO

PPO vs RLOO (usullari)

- RLHF da model savolga javob yozadi
- Reyting hakam bu javobga ball qo'yadi
- Model yuqori ball olgan javoblarni ko'proq yozishni o'rganadi
- Bu yerda asosiy savol: "Bu javob kutilgandan yaxshimi yoki yomonmi?" Bu kattalik "advantage" deb ataladi
- PPO bilan RLOO aynan shu advantageni qanday hisoblashi bilan farq qiladi

PPO - (Proximal Policy Optimization)
RLOO - (Reinforce leave one out)

LLM Post training xulosa

LLM – Katta xajmdagi matnlar asosida o'qitilgan, inson tilini tushunish va generatsiya qilishga mo'ljallangan model

Pre training – Modelni katta hajmdagi ma'lumotlar asosida dastlabki o'qitish jarayoni

Post training – Oldindan o'qitilgan modelni foydalanuvchi ko'rsatmalariga mos, foydali va xavfsiz javob berishga moslashtirish jarayoni

Fine-Tuning – Tayyor modelni maxsus ma'lumotlar yordamida qo'shimcha o'qitish

Alignment – Model javoblarini inson xohishi, ko'rsatmalari va qadriyatlariga moslashtirish

Base Model – Pre training'dan o'tgan, lekin hali maxsus instruction tuning qilinmagan model.

Instruct Model – Inson ko'rsatmalariga javob berishga moslashtirilgan model

checkpoint – Modelning ma'lum bir o'qitish bosqichida saqlangan holati

SFT

SFT – Modelni savol-javob yoki instruction-response ma'lumotlari orqali o'qitish

Instruction Tuning – Modelga berilgan buyruq va topshiriqlarni to'g'ri bajarishni o'rgatish

Training Dataset – Modelni o'qitish uchun tayyorlangan ma'lumotlar to'plami

Instruction – Model bajarishi kerak bo'lgan topshiriq yoki ko'rsatma

Response – Model berishi kerak bo'lgan javob

prompt – Modelga yuborriladigan birinchi matn yoki ko'rsatma

Completion – Model tomonidan generatsiya qilingan matnni davomi yoki javobi

Data Cleaning – O'qitish ma'lumotlaridagi xato, takroriy va kerak siz ma'lumotlarni tozalash

Data Formatting – Ma'lumotni model talab qiladigan formatga keltirish

JSONL – Har bir qator alohida JSON obyektidan iborat ma'lumotlar formati.

PEFT, LoRA va QLoRA

PEFT – Modelning barcha parametrlarini emas, faqat kichik qismini o'z-gartirib fine-tuning qilish usullari to'plami.

LoRA – Modelning mavjud og'irliklarini to'liq yangilamasdan kichik o'zgarishlar matritsalar orqali o'qitish usuli.

QLoRA – Quantization va LoRA kombinatsiyasi, kamroq GPU xotirasida katta modellarni fine-tuning qilish imkonini beradi.

Adapter – Asosiy modelga qo'shiladigan fine-tuning davomida o'rganiladigan kichik parametrlar to'plami.

Rank (r) – LoRA matritsalarining o'lchamini belgilovchi parametr.

Alpha – LoRA yangilanishining kuchini boshqaruvchi parametr.

Dropout – O'qitish davomida ayrim neyronlarni vaqtincha o'chirib turish orqali overfittingni kamaytiruvchi usul.

Merge Adapter – o'rgatilgan lora adapter parametrlarini asosiy modelga birlashtirish

Frothen weights – o'qitish davomida o'zgartirilmaydigan model parametri

Trainable Parameter – o'qitish jarayonida yangilanadigan parametrlar

DPO va Preference Optimization

Preference Learning – Modelga qaysi javob yaxshiroq ekanligini inson tanlovi orqali o'rgatish.

DPO – (Direct preference optimization) – Model inson tanlagan yaxshi va yomon javoblar asosida to'g'ridan-to'g'ri optimallashtirish usuli.

Chosen response – inson tomonidan yaxshi deb tanlangan javob

Rejected response – inson tomonidan yomon deb tanlangan javob

Preference Dataset – Chosen va rejected javoblar juftlaridan tashkil topgan dataset

Reference Model – DPO jarayonida taqqoslash uchun ishlatiladigan boshlang'ich model nusxasi

Reward Model – Model javoblarga sifat yoki afzallik ballarini beruvchi organgan model

Reward Function – Model javobining qanchalik yaxshi ekanligini baholovchi funksiya

Beta (β) – DPO jarayonida yangi model va reference model orasidagi farqni boshqaruvchi parametr

RLHF – inson fikr-mulohazalari asosida modelni optimallashtirish yondashuvi.

PPO – Proximal Policy Optimization – RLHF jarayonida ishlatiladigan reinforcement learning algoritmi.

ORPO – odds ratio preference optimization – preference learning uchun ishlatiladigan optimallashtirish usuli.

Warmup – training boshida LR qiymatini asta sekin oshirish.

Weight decay – Modelni ortiqcha moslashib ketishini kamaytirishga yordam beruvchi regularizatsiya.

Gradient clipping – Juda katta gradient qiymatlarini cheklash usuli.

Token – Model matnni qayta ishlashda foydalanadigan eng kichik birliklardan biri.
Tokenizer – Matnni tokenlarga, tokenlarni esa sonli ID larga aylantiruvchi vosita.
Vocabulary – Tokenizer biladigan barcha tokenlar to'plami.
Embedding – Tokenlarni sonli vektorlarga aylantirish.
Context window – model bir vaqtda qayta ishlay oladigan tokenlar chegarasi.
Transformer – Attention mexanizmiga asoslangan neyron tarmoq turi.
Self-Attention – Matndagi tokenlarni bir-biriga bog'lagan ahamiyatini aniqlash mekanizmi.
Multi Head Attention – Bir nechta Attention mexanizmlarini parallel ishlatish.
Causal Language Modeling – Keyingi token oldingi tokenlar asosida bashorat qilish usuli.
Quantization – model og'irliklarini kamroq bitlarda ifodalash orqali xotira sarfini kamaytirish.

FP32 – 32 bit floating point sonlar formati
FP16 – 16 bit floating point sonlar formati
BF16 – 16 bit aniq floating point sonlar formati
INT8 – 8 bit integer formatida model parametrlarini saqlash
INT4 – 4 bit ixcham formada model parametrlarini saqlash

GGUF – LLM modellarini, ayniqsa llama.cpp ekotizimida, saqlash va ishga tushirish formati

Inference – O'rgatilgan modeldan foydalanib javob olish bosqichi

KV Cache – Attention hisoblashlarida saqlash orqali inference jarayonini tezlatish

Perplexity – Modelning matnni bashorat qilish noaniqligini o'lchovchi metrika

Benchmark – Modellarning natijalarini standart testlar orqali taqqoslash

Hallucination – Modelning noto'g'ri yoki mavjud bo'lmagan ma'lumotni ishonchli ko'rsatishi

Hugging Face – Tayyor modellar, datasetlar va ML vositalari platformasi

Transformers – Hugging Face kutubxonasi, LLM modellarini oson ishlash imkonini beradi

Datasets – Datasetlarni yuklash va tayyorlash kutubxonasi

TRL – SFT, DPO, PPO va boshqa post-training jarayonlarini bajarish uchun kutubxona

PEFT – Lora va boshqa parametr-efficient fine-tuning usullari orqali o'qitish vositasi

Accelerate – model trainingni GPU va bir nechta qurilmalarda boshqarish vositasi

PyTorch – DL modellarini yaratish va o'qitish framework'i

Weights & Biases – Training jarayoni metrikalari va tajribalarni kuzatish platformasi

Tensor Board – Training natijalari va grafiklarini kuzatish vositasi

Unsloth – LLM fine-tuningni xotiradan tejab va tezlatadigan kutubxona

Ollama – LLM modellarini lokal ishga tushirish va boshqarish vositasi

llama.cpp – Quantized LLM modellarini turli qurilmalarda yengil ishga tushiruvchi kutubxona